llmXive Automated Review of Infinite Worlds with Versatile Interactions

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong quantitative and comparative claims that lack direct empirical support in the provided text or tables.

First, Table 1 asserts that the proposed model is the only one with "Infinite" semantic interaction, contrasting it with "Few" for baselines like Genie 3. This binary distinction appears to be a qualitative overstatement unless supported by a specific metric defining the interaction vocabulary size. The text does not clarify if "Few" implies a hard limit or a relative scarcity, making the "Infinite" claim potentially misleading without further evidence.

Second, the central claim of "unbounded interaction horizon" with "no visible decay" over an hour-long session relies entirely on qualitative descriptions and a figure caption. There are no reported quantitative metrics (such as FVD, LPIPS, or a drift score) tracking quality degradation over the 60-minute rollout. Without numerical evidence demonstrating that error accumulation is suppressed to a negligible level, the assertion that stability is "structural" remains an unsupported inference.

Finally, the abstract and introduction state the system "guarantees" 60 fps at 720p. While Section 4 details the optimization stack, the Results section fails to report the actual measured inference speed or latency. A "guarantee" implies a verified bound; the absence of a specific FPS measurement in the results renders this claim unsupported by the paper's own data.

Action items.

- [**writing**] Table 1 claims 'Ours' has 'Infinite' semantic interaction vs 'Few' for Genie 3. Verify if this binary distinction is supported by quantitative data or if it overstates the difference relative to the cited baselines' actual capabilities.
- [**science**] Section 1 and 5 claim 'no visible decay' over an hour-long rollout but provide no quantitative metrics (e.g., FVD, PSNR) to support this. Add a metric or soften the claim to 'qualitatively stable' based on the figure.
- [**science**] The abstract claims a 'guarantee' of 60 fps at 720p, but Section 5 reports no measured FPS or latency. Report the actual throughput or remove the word 'guarantee' to align with the evidence.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 serves as a visual teaser with a collage of generated scenes, but the caption is grammatically incomplete, lacking a subject and proper sentence structure.

Figure 2

Figure 2 is a clear and well-structured pipeline diagram that effectively visualizes the data engine workflow described in the caption. All stages, from data acquisition to the final training dataset, are logically organized with distinct icons and labels that are easy to read.

Figure 3

Figure 3 effectively visualizes the pipeline described in the caption, showing the autoregressive generation of world states from an initial image and user inputs. The diagram clearly maps the flow from the 'Initial State' and 'User Input' blocks through the 'DiT Block' to the resulting '1st State', '2nd State', and '3rd State', with the legend and labels being legible and consistent with the text.

Figure 4

The figure provides a clear visual overview of the DiT block and attention masks, but the 'Cross-Attn Mask' grid contains undefined labels (aG, aB) that contradict the caption's description of a single 'global prompt', and the vertical orientation of the 'Plücker Encoder' text reduces readability.

Figure 5

Figure 5 is a clear and well-structured diagram that effectively visualizes the Agentic Interaction Harness described in the caption. The flow from user input through the VLM 'Director' and Cerebellum 'Pilot' to the updated world state is logical, and the distinction between Mode A and Mode B is visually distinct.

Figure 6

The figure effectively demonstrates the temporal progression of the interactions described in the caption, but the resolution is too low to verify the specific UI elements (floating windows) mentioned, and the text overlays are illegible.

Figure 7

The figure effectively demonstrates prompt switching and object interaction with the office chair, but fails to visually support the caption's claim of 'diverse protagonists' and contains illegible UI elements.

Figure 8

Figure 8 effectively demonstrates the model's capability for seamless prompt switching and world exploration across diverse global landmarks. The visual sequence clearly supports the caption's claim of smooth semantic evolution, and the image quality is sufficient to verify the transitions.

Figure 9

Figure 9 effectively visualizes the interactive system interface described in the caption. The layout clearly distinguishes between the live generation viewport, low-level controls, and the agentic control surface, with all key elements (WASD, IJKL, Event Proposals) correctly labeled and legible.

Figure 10

The figure effectively visualizes a long-horizon rollout, but the caption claims 20 scenarios while only 19 are shown. Additionally, the alternating direction of the timeline arrows across rows is not explained and may hinder the understanding of the temporal sequence.

Figure 11

The figure displays a grid of video frames but lacks any labels or legends to identify the models being compared, rendering the 'qualitative comparison' claim unverifiable. Additionally, the caption is too generic to explain the specific scenarios shown.

Figure 12

The figure presents a qualitative comparison of three video generation models but fails to support the caption's specific claim of evaluating 'causal pretraining' or identifying which model is the proposed method versus the baselines.

Action items.

- [**writing**] Figure 1: The caption is a sentence fragment ('generates infinite worlds...') rather than a complete descriptive sentence.
- [**writing**] Figure 1: The caption lacks a subject (e.g., 'Our model' or 'The system') to clarify what entity is performing the action.
- [**science**] Figure 4: The 'Cross-Attn Mask' grid shows a lower-triangular pattern for the autoregressive component (rows x0-x2 attending to a0-a2), but the caption states this component attends to 'chunk-wise prompts of lower-triangular pattern'. However, the grid also shows the bidirectional component (rows x0t-x2t) attending to aG and aB, but the caption says it attends to a 'global prompt'. The grid labels aG and aB are not defined in the caption or figure, making it unclear if they represent the global
- [**writing**] Figure 4: The 'Plücker Encoder' box is oriented vertically with text rotated 90 degrees, which is difficult to read and inconsistent with other horizontal text elements in the diagram.
- [**writing**] Figure 6: The text overlaying the video frames is illegible due to low resolution and poor contrast, preventing verification of the 'dynamic interactive floating windows' described in the caption.
- [**writing**] Figure 6: The caption describes a 'tracking-mode interface' with 'event cards,' but

the figure lacks a clear UI frame or legend to distinguish the interface elements from the raw video content.

- [**science**] Figure 7: The caption claims 'controllable navigation of diverse protagonists,' but the visual evidence exclusively features an office chair as the protagonist; the 'diverse' aspect is not demonstrated in this specific figure.
- [**writing**] Figure 7: The image contains small, illegible UI icons in the bottom-left corners of the panels that are too blurry to read, obscuring potential control inputs or status indicators.
- [**science**] Figure 10: The caption claims to cover '20 distinct scenarios,' but the figure displays only 19 labeled frames (G01–G18 plus START and FINAL), creating a discrepancy between the text and the visual evidence.
- [**writing**] Figure 10: The timeline arrows in rows B and D point leftward (decreasing time), while rows A and C point rightward (increasing time). This zig-zag layout is not explained in the caption and may confuse the chronological progression of the 'one-hour journey'.
- [**science**] Figure 11: The caption claims 'Qualitative comparisons' but the figure contains no labels, legends, or text identifying which rows correspond to the authors' model versus the baselines, making the comparison impossible to evaluate.
- [**writing**] Figure 11: The caption is generic and does not describe the specific content shown (fire extinguisher and jet ski scenarios), failing to link the visual evidence to the claim of 'stable visual and physical consistency'.
- [**science**] Figure 12: The caption claims 'Qualitative comparisons on causal pretraining' and states the model shows 'stable performance compared with baselines,' but the figure displays three distinct video generation examples (eagle, fire extinguisher, jet ski) comparing MAGI, HY-World 1.5, and Lingbot-World 2.0. There is no visual evidence of 'causal pretraining' (e.g., ablation studies, loss curves, or specific pretraining artifacts) nor a clear baseline comparison demonstrating stability; the figure ap
- [**writing**] Figure 12: The caption is generic ('Our model shows stable performance...') and does not identify which of the three models (MAGI, HY-World 1.5, Lingbot-World 2.0) corresponds to 'Our model' or the 'baselines,' forcing the reader to guess the mapping.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-written and avoids excessive in-group slang, making it accessible to a competent reader from an adjacent field (e.g., a computer vision or NLP researcher). Most acronyms (VLM, DiT, DMD, PF-ODE) are either standard across the broader ML community or defined upon first use.

However, there are a few instances where specific subfield terminology or acronyms are used without immediate expansion, which could cause a momentary stall for a reader not deeply embedded in the specific intersection of video diffusion and world modeling.

1. **Plücker embeddings (Section 3.2):** This is a specific geometric concept. While standard in 3D vision, a reader from NLP or general RL might not know it refers to a 6D ray representation. A brief parenthetical gloss would be helpful.
2. **SAM (Section 4.2):** The text refers to "SAM-based" without explicitly spelling out "Segment Anything Model" in the main body text (it is only in the figure caption). The acronym should be expanded at first use in the prose.
3. **KV Cache (Section 4.3):** "KV" is a common abbreviation for "Key-Value" in transformer literature, but it is not universally known outside that specific sub-ecosystem. Expanding it to "Key-Value (KV) cache" at first use removes ambiguity.
4. **Notation Definition (Section 3.1):** While is standard, explicitly stating "where represents the model parameters" in the sentence introducing Equation 1 or immediately following it ensures no ambiguity for readers from fields where might denote a different quantity (e.g., temperature in some physics contexts, though unlikely here).

These are minor fixes that significantly improve the self-containment of the paper for the target "adjacent-field PhD" audience.

Action items.

- [**writing**] Section 3.1 (Formulation): The symbol is not used, but the symbol is introduced in Equation 1 without explicit definition of its domain (e.g., 'where represents the learnable parameters of the world model'). While standard in ML, explicit definition at first use in the equation block aids adjacent-field readers.
- [**writing**] Section 3.2 (Pre-Training): The term 'Plücker embeddings' is used without a brief gloss (e.g., 'a six-dimensional representation of 3D rays'). While known in computer vision, an adjacent-field PhD in NLP or general ML may not immediately recall the specific geometric encoding.
- [**writing**] Section 3.2: The acronym 'MoBA' (Mixture of Bidirectional and Autoregressive) is introduced and defined, but the text later refers to 'MoBA mask' and 'MoBA' interchangeably. Ensure the first use of the acronym is immediately followed by the full name in the same sentence or clause for clarity.
- [**writing**] Section 4.2 (Agentic Interaction Harness): The term 'SAM-based' is used. While 'Segment Anything Model' is a well-known benchmark, the acronym 'SAM' is not explicitly expanded in the text (it appears in the caption of Fig 2 but not the prose). Expand 'SAM' to 'Segment Anything Model (SAM)' at first use in Section 4.2.
- [**writing**] Section 4.3 (Visual Quality Enhancement): The term 'KV Cache' is used without

expansion. While standard in LLM/DiT literature, an adjacent-field reader might not know 'KV' stands for 'Key-Value'. Expand to 'Key-Value (KV) cache' at first use.

- **[writing]** Section 5.1: The phrase 'causal distilled' is used as a compound adjective. While understandable, it is slightly non-standard shorthand. Consider 'causally distilled' or 'distilled causal model' for grammatical precision, though this is a minor stylistic point.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper's argument structure is generally sound, with a clear progression from the problem of long-horizon instability to the proposed causal pretraining and distillation solution. However, there are two specific instances where the text's conclusions do not strictly follow from the presented evidence or contradict other sections of the paper.

First, in Section 5.1 ("Comparison with prior work"), the authors state: "it is the only system in our comparison that sustains high-resolution generation in real time without visible degradation." This claim of exclusivity directly contradicts Table 1 ("Comparison with recent interactive world models"), where the "Real-time" row contains checkmarks for M-G 3.0, D-W, LingBot-World, and HappyOyster. If the table is accurate, the text's claim that the authors' model is the *only* real-time system is false. If the table is inaccurate (i.e., those baselines are not truly real-time), the table must be corrected. As written, the text draws a conclusion that is not entailed by the provided data table.

Second, Table 1 lists the "Semantic Interaction" capability of the proposed model ("Ours") as "Infinite." This terminology is logically inconsistent with the rest of the paper. The Abstract, Introduction, and Results sections describe the action space as "rich," "versatile," and comprising a "broader spectrum of actions" (e.g., attacking, archery, spell-casting). These are finite, discrete categories. Describing a finite set of actions as "Infinite" is a category error that contradicts the specific examples given in the text. The term "Infinite" is correctly applied to the "Generation Duration" (unbounded time), but misapplied to the interaction space.

These issues are primarily matters of precise wording and data consistency rather than fundamental flaws in the causal logic of the method itself. Correcting the text to match the table (or vice versa) and refining the terminology in the table will resolve the logical gaps.

Action items.

- **[writing]** Section 5.1 claims the model is the 'only system... that sustains high-resolution generation in real time,' but Table 1 lists 'M-G 3.0', 'D-W', 'LingBot-World', and 'HappyOyster' as having the 'Real-time' checkmark. The text's exclusivity claim contradicts the table's data. Either remove the 'only' qualifier or correct the table's 'Real-time' column for the baselines.

- **[writing]** Table 1 claims 'Semantic Interaction' for 'Ours' is 'Infinite,' while the text in Section 1 and 5 describes a 'rich, controllable action space' and 'versatile interactions.' The term 'Infinite' for a discrete set of actions (combat, archery, etc.) is logically imprecise and contradicts the finite nature of the described action vocabulary. Change 'Infinite' to 'Rich' or 'Diverse' to match the text.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several sweeping claims regarding the scope and capability of the system that exceed the evidence provided in the evaluation section.

First, the title, abstract, and introduction repeatedly use the terms "infinite" and "unbounded" to describe the interaction horizon and world generation (e.g., Abstract: "unbounded interaction horizon"; Intro: "unbounded, drift-free interactive world"). The only evidence provided for this claim is a single qualitative figure (Fig. 5, "Hour-long world rollout") showing a 60-minute session. While impressive, a 60-minute demonstration does not constitute evidence of "infinite" or "unbounded" stability. The paper does not provide quantitative metrics (e.g., drift error accumulation curves) extending beyond this single hour, nor does it test the system at significantly longer horizons (e.g., 10+ hours) to justify the "infinite" descriptor. The claim should be narrowed to "extended (hour-scale) stability" or supported by quantitative analysis of error accumulation over time.

Second, Table 1 and the abstract claim the model supports "Infinite" semantic interaction. The evaluation section (Sec 5) and the introduction list a specific set of actions (combat, archery, spell-casting, shooting) and environmental changes. While the action space is described as "rich" and "diverse," the term "Infinite" suggests a capability to handle an unbounded or open-ended set of semantic inputs that has not been empirically tested. The evidence supports a claim of "expanded" or "versatile" interaction, but not "infinite."

Third, the abstract states the system "guarantees rapid response time" sufficient for 720p at 60 fps. The word "guarantees" implies a hard engineering constraint or a worst-case bound that is not demonstrated. The paper reports achieved performance on specific hardware (implied by the "single GPU" mention for the 1.3B model, though the 14B model's deployment requirements are less specific). Without a formal latency analysis or a guarantee of performance across varying loads, "guarantees" is an overreach. "Achieves" or "demonstrates" would be more accurate.

Finally, while the paper includes a "Limitations" section, it focuses on memory, identity consistency, and physics. It does not explicitly acknowledge the limitation that the "infinite" and "unbounded" claims are currently only validated for a single hour-long session, which is a critical boundary for the paper's central contribution. Adding a sentence to the limitations section clarifying that the "infinite" claim is currently supported only by hour-scale qualitative evidence would align the rhetoric with the demonstrated scope.

Action items.

- [**writing**] Abstract claims 'unbounded interaction horizon' and Section 1 claims 'infinite worlds,' but evidence is limited to a single hour-long qualitative demo (Fig 5). Replace 'unbounded/infinite' with 'extended (hour-scale)' or provide quantitative metrics proving stability beyond the tested hour.
- [**writing**] Table 1 and Abstract claim 'Infinite' semantic interaction, yet the evaluation (Sec 5) only demonstrates a specific set of actions (combat, archery, etc.) in a curated demo. 'Infinite' implies an unbounded vocabulary or capability never tested; narrow to 'rich and diverse' or 'expanded action space'.
- [**writing**] Abstract states the system 'guarantees rapid response time' for 720p/60fps. 'Guarantees' is a strong engineering claim not supported by the text, which only reports achieved throughput on specific hardware. Change to 'achieves' or 'demonstrates' and specify the hardware configuration.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a generative world model capable of real-time, infinite-horizon video synthesis with interactive control. From a safety and ethics perspective, the work does not exhibit the specific, non-trivial risks that would require mitigation or disclosure beyond standard practice for this subfield.

The data pipeline (Section 2) relies on a mix of self-collected egocentric videos, synthetic data from game engines (Unreal Engine), and large-scale web videos. The paper cites standard public datasets (e.g., Ego4D, EPIC-Kitchens) and does not claim to use private, sensitive, or non-public human-subject data requiring IRB approval. The use of web-scraped video is a common practice in the field; while the paper does not detail the specific licensing of every web source, it does not release a raw scraped dataset that would violate redistribution terms, nor does it claim to have bypassed specific Terms of Service in a way that creates a unique liability for this specific release. The synthetic data component further mitigates concerns regarding human consent.

The system's capabilities (combat, spell-casting, environmental manipulation) are framed within a simulated, generative context. While the "agentic harness" allows for autonomous behavior, the paper does not describe a system designed for real-world deception, surveillance, or the generation of actionable cyber/physical exploits. The "combat" and "shooting" actions are visual simulations within a generated world, not instructions for real-world harm. The paper does not provide operational details for generating harmful content (e.g., biological agents, specific malware) nor does it claim to have discovered a vulnerability in a live system that requires responsible disclosure.

There is no evidence of Personally Identifiable Information (PII) being released in the dataset or figures. The authors acknowledge limitations regarding long-term memory and physical consistency, but these are technical constraints rather than safety failures. As this is a third-party preprint, the absence of a formal "Broader Impacts" statement is noted but does not constitute a fatal flaw given the low-risk nature of the methodology and the standard norms for video generation papers. The work is a standard contribution to the field of generative world models with no foreseeable, unmitigated safety risks identified in the text.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling system for infinite-horizon interactive world modeling, but the evidentiary support for its central quantitative claims is currently insufficient to rule out alternative explanations such as cherry-picked demonstrations or baseline under-tuning.

The most significant gap lies in the comparison of generation duration. Table 1 and Section 5.1 assert that the proposed model achieves "Hours (Infinite)" generation while competitors are limited to "Minutes." This is a massive qualitative leap that requires robust quantitative backing. Currently, the evidence consists of a single hour-long qualitative rollout (Figure 5) and visual comparisons. A skeptical reader cannot determine if the baselines (e.g., Genie 3, HappyOyster) were simply not evaluated for long horizons, or if the proposed model's stability is a result of a specific, favorable prompt rather than a structural property. To support the "infinite" claim, the authors must report a standardized long-horizon benchmark (e.g., 30-60 minutes) with performance metrics (such as FID, CLIP score, or a specific drift metric) averaged over multiple random seeds for both their model and the baselines. A single anecdotal run is not sufficient evidence for a claim of this magnitude.

Furthermore, the attribution of success to specific architectural components (MoBA mask, DMD loss) lacks isolation. The paper claims these components prevent drift, yet the ablation studies are missing. The observed stability could plausibly arise from the increased training data, longer training duration, or the specific distillation schedule rather than the proposed attention masks or loss functions. The authors need to provide a quantitative ablation table showing the drift metrics for the full model versus variants where the MoBA mask is replaced with standard causal masking, and where the DMD loss is removed, while holding compute and data constant.

Finally, the "real-time" claim (60 fps) is presented as a definitive capability but lacks the necessary statistical rigor. The paper describes system optimizations but does not provide a latency distribution or sustained throughput measurements over a long session. Real-time performance in interactive systems is often defined by the tail latency (p99), not just the mean. Without reporting the distribution of frame generation times and the specific hardware configuration used to achieve 60 fps, the claim remains an unverified assertion.

In summary, while the qualitative results are impressive, the paper requires quantitative validation of its core claims regarding duration, component efficacy, and real-time performance to be considered scientifically sound.

Action items.

- [**science**] The paper presents a compelling system for infinite-horizon interactive world modeling, but the evidentiary support for its central quantitative claims is currently insufficient to rule out alternative explanations such as cherry-picked demonstrations or baseline undertuning. The most significant gap lies in the comparison of generation duration. Table 1 and Section 5.1 assert that the proposed model achieves "Hours (Infinite)" generation while competitors are limited to "Minutes." This is a ma

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The paper presents a system for infinite-horizon interactive world modeling but lacks the statistical rigor required to validate its quantitative claims. Section 5.1 asserts that the proposed model "matches or exceeds" baselines and is the "only system" to achieve real-time, infinite generation. These claims are supported exclusively by qualitative figures (Fig. 1, Fig. 4) and a single hour-long demo, with no reported numerical metrics (e.g., FVD, PSIM, throughput in fps) or measures of uncertainty (standard deviation, confidence intervals). In the absence of quantitative data, the claim of superiority is anecdotal.

Specifically, the "long-horizon stability" claim relies on visual inspection of a single 60-minute rollout. Without a quantitative drift metric (e.g., decay in feature similarity over time) reported with variance across multiple seeds, it is impossible to distinguish structural stability from a favorable single run. Similarly, Table 1 uses binary checkmarks for "Real-time" and "Generation Duration" without providing the underlying throughput numbers or duration measurements with error bars. To meet statistical standards, the authors must report mean performance metrics with standard deviations over multiple independent runs (seeds) for all baselines and the proposed method, and provide a quantitative analysis of drift over time rather than relying on qualitative "no visible decay" assertions.

Action items.

- [**writing**] The paper presents a system for infinite-horizon interactive world modeling but lacks the statistical rigor required to validate its quantitative claims. Section 5.1 asserts that the proposed model "matches or exceeds" baselines and is the "only system" to achieve real-time, infinite generation. These claims are supported exclusively by qualitative figures (Fig. 1, Fig. 4) and a single hour-long demo, with no reported numerical metrics (e.g., FVD, PSIM, throughput in fps) or measures of uncertai

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured and readable, with a clear logical flow from problem statement to method and results. The abstract effectively summarizes the four key upgrades. However, there are several specific areas where the prose can be tightened to improve clarity and consistency.

First, there is a terminological inconsistency in Section 2. The introduction lists "multi-granularity annotation" as the third stage, but the corresponding subsection is titled "Multi-dimensional Video Annotation." This shift in terminology ("granularity" vs. "dimension") is confusing and should be standardized to ensure the reader understands these are the same concept.

Second, Section 3 contains a slightly ambiguous sentence regarding the "Mixture of Bidirectional and Autoregressive Attention Mask" (MoBA). The phrase "promoting its transition from bidirectional to autoregressive generation" could be misinterpreted as the model switching modes entirely, rather than balancing the two. Clarifying that the mechanism facilitates a *balance* or *smooth transition* between these modes would prevent confusion.

Third, in Section 5, the phrase "supports much more interactions" is grammatically awkward and imprecise. It should be rephrased to "supports a significantly broader range of interactions" to match the formal tone of the rest of the paper.

Finally, the "Limitations" section in Section 6 lists four distinct points but introduces them with a single sentence ("Several limitations remain.") followed immediately by "The most fundamental is...". Adding ordinal markers (e.g., "First," "Second,") or a brief introductory phrase for each point would improve the paragraph's rhythm and readability.

Overall, these are minor stylistic and consistency issues that do not impede understanding but, if addressed, would make the paper more polished.

Action items.

- [**writing**] The paper is generally well-structured and readable, with a clear logical flow from problem statement to method and results. The abstract effectively summarizes the four key upgrades. However, there are several specific areas where the prose can be tightened to improve clarity and consistency. First, there is a terminological inconsistency in Section 2. The introduction lists "multi-granularity annotation" as the third stage, but the corresponding subsection is titled "Multi-dimensional Video An